

Linux Essentials 010–160;

Key Terms & Commands to Recognize

Topic 1: The Linux Community and a Career in Open Source

- **Package Managers:** * dpkg, apt-get — Used for Debian-based distributions (Ubuntu, Debian).
 - rpm, yum (and dnf) — Used for Red Hat-based distributions (RHEL, Fedora, CentOS).
- **Key Software to Recognize:** * *Desktop:* LibreOffice, Firefox, Thunderbird, GIMP.
 - *Server/Cloud:* Apache HTTPD, NGINX, MariaDB, MySQL, NFS, Samba, Nextcloud.
 - *Languages:* Python, PHP, Java, JavaScript, Perl, C.

Topic 2: Finding Your Way on a Linux System

2.1 Command Line Basics

- echo — Prints text or variable values to the screen (e.g., echo \$PATH).
- history — Displays a list of previously executed commands.
- export — Sets or changes an environment variable (e.g., export MYVAR="value").
- type — Identifies how a command will be interpreted (e.g., alias, built-in shell command, or executable file).
- *Key Concept (The PATH variable):* \$PATH is a colon-separated list of directories where the shell looks for executable files.

2.2 Using the Command Line to Get Help

- man — Displays the official manual page for a command (e.g., man ls). Use man -k or apropos to search by keyword.
- info — Provides an alternative, hyperlink-style menu-driven documentation layout.
- /usr/share/doc/ — The directory where supplemental documentation and text readmes for installed packages live.

2.3 Using Directories and Listing Files

- pwd — **P**rint **W**orking **D**irectory. Shows exactly where you currently are in the filesystem.
- cd — **C**hange **D**irectory.
 - cd ~ or just cd — Goes to your home directory.
 - cd .. — Moves up one directory level.
 - cd . — Refers to the current directory.
- ls — **L**ist directory contents.

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- `ls -a` — Shows all files, including hidden files (files starting with a dot `.`).
- `ls -l` — Shows a long detailed listing (permissions, owner, size, timestamps).
- `ls -R` — Recursively lists subdirectories.
- `ls -h` — Human-readable sizes (e.g., 4K, 23M).

2.4 Creating, Moving, and Deleting Files

- `mkdir` — Makes a new directory. (Use `-p` to create nested parent directories).
- `rmdir` — Removes an *empty* directory.
- `touch` — Creates a new empty file or updates timestamps of an existing file.
- `cp` — Copies files and directories. (Use `-r` to copy recursively).
- `mv` — Moves or renames files and directories.
- `rm` — Removes (deletes) files. (Use `-r` to delete directories, `-f` to force).

Topic 3: The Power of the Command Line

3.1 Archiving Files on the Command Line

- `tar` — **T**ape **A**rchive tool used to group files together.
 - `tar -cvf archive.tar files/` — **C**reate a file.
 - `tar -xvf archive.tar` — **E**xtract a file.
 - `tar -tvf archive.tar` — **T**est/List contents.
 - `tar -zcvf archive.tar.gz files/` — Compress using **gzip**.
 - `tar -jcvf archive.tar.bz2 files/` — Compress using **bzip2**.
 - `tar -Jcvf archive.tar.xz files/` — Compress using **xz**.
- **Compression stand-alones:** `gzip/gunzip`, `bzip2/bunzip2`, `xz/unxz`.

Compression program	Uncompression program	Filename extension
<code>gzip</code>	<code>gunzip</code>	<code>.gz</code>
<code>bzip2</code>	<code>bunzip2</code>	<code>.bz2</code>
<code>xz</code>	<code>unxz</code>	<code>.xz</code>

- `zip / unzip` — Standard multi-file compression utilities compatible with Windows environments. Compression levels 0 – 9 (*maximum*).

3.2 Searching and Extracting Data from Files

- `cat` — Concatenates and displays file contents entirely.

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- **less** — Allows interactive page-by-page viewing of long text files.
- **head** — Shows the first 10 lines of a file (e.g., `head -n 5 file.txt`).
- **tail** — Shows the last 10 lines of a file (e.g., `tail -f file.log` follows logs in real-time).
- **sort** — Sorts lines of text alphabetically or numerically (`sort -n`).
- **cut** — Extracts sections/columns from lines of text (e.g., `cut -d: -f1 /etc/passwd`).
- **wc** — **W**ord **C**ount. Counts lines (-l), words (-w), or bytes (-c).
- **regex** – regular expressions;

Operator	Description
^	The circumflex symbol marks the beginning of a pattern (line), although in some cases it can be omitted.
\$	Same as with the circumflex symbol, the dollar sign marks the end of a search pattern (line).
.	The period matches any single character.
?	The question is optional and will be matched, at most, once.
+	The plus sign matches the preceding pattern one or more times.
*	The asterisk matches up zero or more times.
 	The pipe symbol indicates that a match can be one of the two terms on either side of the pipe.

- **grep** — Searches for global regular expression patterns within files.
 - `grep -i` — Ignore case.
 - `grep -v` — Invert match (show lines that do *not* match).
- **Redirection / Pipes:**
 - `>` — Redirects standard output to a file (overwrites file).
 - `>>` — Redirects standard output to a file (appends to file).
 - `2>` — Redirects std error (error messages) to a file.
 - `|` — Pipe symbol. Feeds the output of one command as input into another (e.g., `ls -l | grep txt`).

3.3 Turning Commands into a Script

- **Editors:** `vi` / `vim` and `nano`.
- **Script Basics:** Scripts require a "Shebang" line at the very top: `#!/bin/bash`.

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- **Execution:** A script must have executable permission (`chmod +x script.sh`) and is run using `./script.sh`.

Topic 4: The Linux Operating System

4.3 Where Data is Stored (Monitoring and Logs)

- `ps` — Shows current snapshot of running **P**rocesses. (`ps aux` or `ps -ef`).
- `top` — Displays dynamic, real-time running system processes and CPU usage.
- `free` — Displays system memory (RAM) and Swap usage.
- `dmesg` — Prints the kernel ring buffer messages (useful for hardware diagnostic tracing).
- *Key Directories:*
 - `/etc/` — System-wide configuration files.
 - `/var/log/` — System logs (messages, syslog, auth logs).
 - `/boot/` — Files needed to start up the system (kernel, GRUB bootloader).
 - `/proc/` — Virtual directory exposing kernel and process states.
 - `/dev/` — Hardware device files (e.g., `/dev/sda` for a drive).
 - `/sys/` — Virtual directory exposing hardware and driver paths.

4.4 Your Computer on the Network

- `ip addr show` (or older `ifconfig`) — Displays network interfaces and assigned IP addresses.
- `ip route show` (or older `route`) — Displays the system routing table and default gateway.
- `ping` — Sends ICMP echo requests to verify remote network connectivity (e.g., `ping 8.8.8.8`).
- `host` (or `dig`) — Queries DNS servers to look up domain name mappings (e.g., `host google.com`).
- `ss` (or older `netstat`) — Displays active network connections and listening ports.
- *Network Files:*
 - `/etc/resolv.conf` — Configures the DNS nameservers.
 - `/etc/hosts` — Static IP-to-hostname mappings lookup file.

Topic 5: Security and File Permissions

5.1 & 5.2 User Management

- `su` — **S**ubstitute **U**ser. Switches your terminal session to another user (usually root: `su -`).

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- `sudo` — Executes a single command with administrative/root privileges.
- `useradd` — Creates a brand new system user account.
- `groupadd` — Creates a new group.
- `passwd` — Changes a user's password.
- *Authentication Files:*
 - `/etc/passwd` — Stores user account info (UID, GID, home directory, shell choice).
 - `/etc/shadow` — Stores securely hashed user passwords.
 - `/etc/group` — Stores group names and user memberships.

5.3 Managing File Permissions and Ownership

- `chown` — **Change Owner**. Changes owner and/or group of a file (e.g., `chown user:group file.txt`).
- `chmod` — **Change Mode**. Modifies access permissions for User (**u**), Group (**g**), and Others (**o**).
 - *Symbolic*: `chmod u+x script.sh` (gives user execute permission).
 - *Numeric*: Reads are 4, Writes are 2, Executes are 1.
 - `chmod 755 file ->` User: `rwX` (4+2+1), Group: `r-X` (4+1), Others: `r-X` (4+1).
 - `chmod 644 file ->` User: `rw-` (4+2), Group: `r--`, Others: `r--`.

5.4 Special Directories and Files

- `ln -s` — Creates a **symbolic link** (shortcut) pointing to a target file.
- `ls -d` — Lists a directory itself instead of its contents.
- *Key Security Directories/Permissions:*
 - `/tmp/` and `/var/tmp/` — Temporary directories anyone can write to.
 - **Sticky Bit (t)**: Indicated by a `t` at the end of directory permissions (like `/tmp`). It prevents users from deleting files owned by other users within that folder.